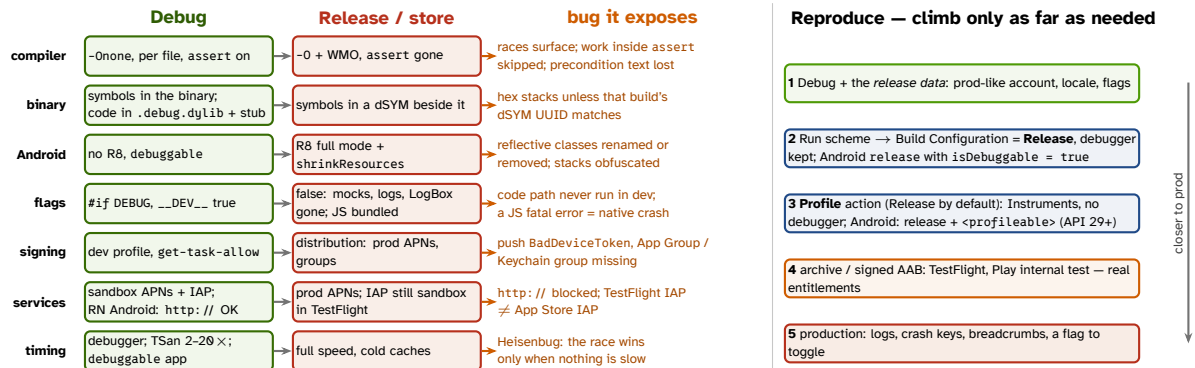


In one line: A Release build is a **different program in a different environment**: optimised code (no asserts, different interleavings, different *observed* object lifetimes), shrunk bytecode (R8), no dev flags (DEBUG, __DEV__), distribution signing with **production** entitlements and services, and no debugger slowing every thread. A bug that “only happens in prod” is a hidden assumption that one of these flips.



Why the program changes

- Optimiser:** -O (Release default; whole-module optimisation on for new projects since Xcode 8) inlines across files, devirtualises, drops retain/release pairs. A data race is a bug at every level yet “can often seem to work” — a faster build changes the interleaving. Swift 6 mode rejects races at compile time; TSan finds the rest (Simulator/macOS only, 2-20x slower). Swift's own docs call -Osize “meant for most production code”; Xcode still defaults Release to -O.
- Lifetimes:** an object is guaranteed only until its *last use*; observed lifetimes are an emergent property of the optimiser (WWDC21). Since **Swift 5.7** (Xcode 14) a named variable is anchored to its scope: its end cannot move across a *deinit barrier* (syncs, weak/unowned loads, pointer access, global or class-member access) — so the classic “weak ref to a local” no longer needs withExtendedLifetime. Code that depends on *when* a deinit runs stays fragile. ObjC ARC locals have *imprecise* lifetime unless marked objc_precise_lifetime.
- Exclusivity:** “Simultaneous accesses to ...”, but modification requires exclusive access” traps in Release too since Swift 5 (Debug only in Swift 4).
- C / ObjC:** an uninitialised local is indeterminate — whatever was there, which changes with the optimisation level. ASan does *not* detect it; MSan runs only on Linux and the BSDs. Use -Wuninitialized, the analyzer, or -ftrivial-auto-var-init=pattern.
- R8** (full mode by default since AGP 8.0) inlines, merges classes, renames, and removes what it cannot see used. Reflection (Gson, Class.forName, ::class.constructors) then throws ClassNotFoundException, NoSuchFieldException ... — add keep rules or @Keep, or use codegen (Moshi codegen, Kotlin Serialization). Names built for getIdentifier(): tools:keep.
- RN:** release embeds the bundle (dev=false) as Hermes bytecode compiled at build time (Metro minify off). No Dev Menu, no LogBox: a fatal JS error is a native crash. Dev mode is *slower* — measure in release: npm run android -- --mode="release", iOS --mode="Release".

Swift checks by optimisation level

	-Onone (Debug)	-O (Release)	-Ounchecked
assert, assertionFailure	trap + message	not evaluated / no effect	assumed true / unreachable
precondition, preconditionFailure	trap + message	trap; message replaced by “precondition failure”	assumed true / unreachable
fatalError + overflow	trap + message trap	trap + message trap	trap + message not checked

Example

```
assert(cache.warmUp()) // -O: never runs
let ok = cache.warmUp(); assert(ok) // fixed
precondition(i < n, "i=\(i) n=\(n)") // -O: message dropped

# Gson reads fields by reflection (Android's conditional rule)
-if class ** { @com.google.gson.annotations.SerializedNameName
    <fields>; }
-keep class <1> { @com.google.gson.annotations.SerializedNameName
    <fields>; <init>(...) } ;
-whyareyoukeeping class com.example.Report # why kept?
```

Which backend each build talks to

build	APNs (aps-environment)	in-app purchase	debugger
Xcode Run, dev profile	development	sandbox (.xcode with a .storekit file)	attaches: get-task-allow
TestFlight	production	sandbox , always	no
App Store	production	production	no

Hosts: api.sandbox.push.apple.com / api.push.apple.com; a token works on one only, the other answers 400 BadDeviceToken.
Transaction.environment (iOS 16): .production / .sandbox / .xcode.

Symptom → likely cause → fix

symptom (Release only)	likely cause	fix
random crash / wrong value under load logic skipped silently	data race; -O changed the interleaving work inside assert, or a #if DEBUG branch	Swift 6 mode; TSan; actor / lock move work out; diff the flags
Android JSON fields null, ClassNotFound push works from Xcode, not TestFlight App Group / shared Keychain empty API calls fail, dev fine	R8 renamed / removed reflective classes token is <i>development</i> ; TestFlight uses production APNs entitlement missing in the distribution profile http://: RN's Android debug build allows cleartext; Android 9+ blocks it; ATS applies in every config TestFlight IAP is sandbox ; server checks the wrong env fatal JS error → RCTFatal (iOS), JavaScriptException (Android)	keep rule, @Keep, codegen store the env with the token regenerate profile; codesign -d HTTPS; scoped exceptions branch on environment global handler + error boundary

Find the flip — diff the two builds

- xcodebuild -showBuildSettings -configuration Debug vs Release: SWIFT_OPTIMIZATION_LEVEL, SWIFT_COMPILATION_MODE, SWIFT_ACTIVE_COMPILATION_CONDITIONS, ENABLE_TESTABILITY, ENABLE_NS_ASSERTIONS, per-config Info.plist.
- codesign -d --entitlements - --xml on the *archived* app.
- Android: diff the build types; apkalyzer manifest debuggable app.apk on the shipped file; merged R8 rules are in build/outputs/mapping/configuration.txt; mapping.txt is overwritten every build — keep one per release.

Traps

- ! A print or a breakpoint makes it vanish — a timing bug. Use signposts/logs, TSan, or a stress loop, not the debugger.
- ! -Onone in Release to “fix” it — hides the race, slower app.

Remember

Compiler · binary · R8 · flags · signing · services · timing — seven things flip; find which one your bug depends on.